Machine Learning

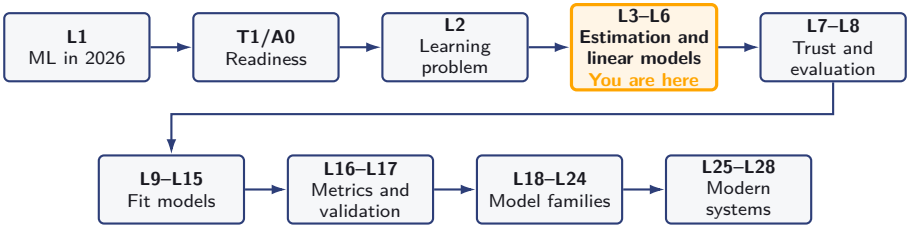
Lecture 3: Estimating Our First Parameters

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Semester I, 2026



Course Roadmap: You Are Here



Lecture 3 estimates our first parameters. Data arrives from a distribution whose numbers we do not know; maximum likelihood is how we guess them, and the recipe does not care which distribution it is.

Where We Got To Last Time

From Lecture 2:

- learning = four choices: data, model, objective, algorithm
- arg min, and convexity
- types and dimensions as a correctness check

Warning

We named a *model* but never said where its numbers come from. Today: they are **estimated**, and here is how.

The Estimation Problem

Two Symbols, For The Rest Of The Course

	What it is	Today
x	the data. *Handed to you.*	40 recorded rainfalls
θ	the parameters. *You must guess.*	the centre and spread

Insight

Every lecture from here is one sentence: **we are handed x, and we must guess θ.** Today we do it for the first time.

The Setup: Data Comes From Somewhere

We assume each observation is **drawn from a distribution** with unknown numbers in it:

$$x_1, x_2, \ldots, x_n \quad \text{drawn independently from} \quad p(x \mid \theta).$$

We know	We do not know	We want
the *shape* of p	the *numbers* θ in it	a good guess $\widehat{\theta}$

Warning

Two assumptions, both real: the observations are **independent**, and the shape we chose is **right**. Neither is free, and we return to both at the end.

What A Distribution Is

A distribution is a rule that says **how often each value shows up**. Nothing more.

Term	Means	In our case
support	the values allowed	rainfall ≥ 0
parameter	a number *inside* the rule	$\theta = (\mu, \sigma)$
normalisation	the total must be 1	the curve encloses area 1

Where they come from. Not from theory first. Somebody measured the same kind of thing many times under the same conditions, drew a histogram, and a shape appeared.

Today's choice, spelled out. Deciding "flood rainfall is bell-shaped" *is* the modelling step. It fixes the formula and names $\theta = (\mu, \sigma)$ — without yet saying what either one is.

Insight

In machine learning the distribution *is* the model. Picking the **shape** — which formula, with its letters still blank — is the modelling decision. Filling in the letters is the arithmetic that follows.

Two Kinds: PMF And PDF

Discrete — a *probability mass function*:

$$p(x) = \Pr(X = x), \qquad \sum_{x} p(x) = 1.$$

Each $p(x)$ is itself a probability, so $0 \le p(x) \le 1$.

Continuous — a *probability density function*:

$$\Pr(a \le X \le b) = \int_a^b p(x)\,dx, \qquad \int p(x)\,dx = 1.$$

Check both totals.

Warning

A density is not a probability. Only its *area* is. This is the single most common confusion in the rest of the course.

The Ones You Will Actually Meet

Name	Models	Parameters	Support	Mean
Bernoulli	one yes/no	θ	$\{0, 1\}$	θ
Binomial	yeses in n tries	n, θ	$0..n$	$n\theta$
Categorical	one of K labels	$\theta_1..\theta_K$	$1..K$	—
Poisson	counts per interval	λ	$0, 1, 2, \ldots$	λ
Normal	a spread of measurements	μ, σ^2	$\mathbb{R}$	μ
Exponential	waiting time	λ	$x \geq 0$	$1/\lambda$
Uniform	no reason to prefer	a, b	$[a, b]$	$(a + b)/2$
Beta	a proportion	a, b	$[0, 1]$	$a/(a + b)$

Insight

The top block is discrete (pmf), the bottom continuous (pdf). Four of these appear later today: normal, Bernoulli, Poisson, exponential — and Beta returns as a prior.

Why These Shapes Keep Reappearing

Assume this	You get this
many small independent effects added up	normal
rare events, many chances	Poisson
waiting, with no memory of how long already	exponential
one trial, two outcomes	Bernoulli

The shapes are not conventions. Each is what you are forced into once you accept an assumption about the mechanism.

Insight

And the choice has teeth: assume normal noise and the natural objective turns out to be $(y - \widehat{y})^2$; assume Bernoulli and it turns out to be cross-entropy. Lecture 4 derives the first of those.

Today's Question

The data

The forty flood days just named. Somebody recorded the rainfall on each, and they scatter around $120\,\mathrm{mm}$, some higher, some lower.

Question

Where is the curve *centred*? And how wide is it? Nobody will tell us — we have only the forty numbers.

Insight

Guessing badly is easy. What we need is a rule that says *which guess is best*, and can defend the answer.

Maximum Likelihood For The Normal

Reading A Normal Curve

A normal density is a bell with a **centre** μ and a **spread** σ:

$$p(x \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\tfrac{1}{2}z^2\right], \qquad z = \frac{x-\mu}{\sigma}.$$

spreads away, $\lvert z \rvert$	0	1	2	2.5
$\exp(-z^2/2)$	1	0.61	0.14	0.044

Insight

Distance from the centre only means something *relative to the spread*. Here $\theta = (\mu, \sigma)$ — the two numbers we must guess.

Step 1: How Surprising Is *One* Reading?

Pick a candidate θ. The curve then tells you a height for a single observation:

$$p(x_i \mid \theta).$$

Large height: this θ finds x_i ordinary.
Small height: this θ finds x_i surprising.
One reading, $x_i = 90$, guess $\mu = 120,\ \sigma = 30$. Evaluate it.

Insight

Read it *backwards*. Normally x varies and θ is fixed. Here x **is fixed — it already happened — and** θ **is what we vary.**

Step 2: All n Readings At Once

Independent observations **multiply**:

$$L(\theta) \;=\; p(x_1 \mid \theta) \times p(x_2 \mid \theta) \times \cdots \times p(x_n \mid \theta) \;=\; \prod_{i=1}^{n} p(x_i \mid \theta).$$

This is the **likelihood**: one number for how well a candidate θ fits the whole dataset.

Warning

It is a function of θ, not of x. It is *not* a probability of θ — it does not add up to one over θ.

Step 3: Take Logs

In Class

Logs turn a product into a sum, and do not move the peak.

$$\ell(\theta) \;=\; \log L(\theta) \;=\; \sum_{i=1}^{n} \log p(x_i \mid \theta).$$

log is increasing, so it preserves *where* the maximum is.
Products become sums: 10^{-80} becomes -184.
Expand $\log p(x_i \mid \mu, \sigma)$ for the normal.

Summary

$\widehat{\theta} = \arg\max_\theta \ell(\theta)$ is the **maximum likelihood estimate**.

Try it: `ml2026.floodresq.com/explore/mle` — move a guess and watch the score, panel 1

Step 4: Find The Peak — The Centre

With σ held fixed, the log-likelihood is

$$\ell(\mu) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Only the second term contains μ.

Differentiate, set to zero, solve.

Summary

The maximum likelihood estimate of the centre is the **sample mean**.

Step 4 Again, For The Spread

Write $v = \sigma^2$ and differentiate with respect to v:

$$\ell(v) = -\frac{n}{2}\log(2\pi v) - \frac{1}{2v}\sum_i (x_i - \widehat{\mu})^2.$$

Differentiate, set to zero, solve — as before.

Warning

Note the $1/n$, not $1/(n-1)$. This estimate of the spread is slightly too small. L8 says why.

On Our Forty Days

The data, again

The **forty flood days** from the start of the lecture: somebody recorded the rainfall on each one, and they scatter around $120\,\mathrm{mm}$. We were never told where their curve sits — that is what we are working out.

centre: $\widehat{\mu} = \frac{1}{n}\sum x_i$

spread: $\widehat{\sigma} = \sqrt{\frac{1}{n}\sum (x_i - \widehat{\mu})^2}$

Add the forty and divide by forty. Then do the same for the squared deviations.

Insight

The forty days give $\widehat{\mu} \approx 118\,\mathrm{mm}$ and $\widehat{\sigma} \approx 31\,\mathrm{mm}$. Rounded, that is the "centred 120, spread 30" we have been using. It was never handed down; it was estimated from recorded days, exactly like this — and real estimates do not arrive round.

Try it: `ml2026.floodresq.com/explore/mle` — the same forty points, panel 1

The Method Beyond The Normal

None Of The Steps Used The Normal

Look back at the four steps. Not one of them used the normal:

1	write $p(x_i \mid \theta)$ for one observation
2	multiply over the n observations
3	take logs
4	differentiate, set to zero, solve for θ

Swap the distribution and only line **1** changes:

Data looks like	**Use**	**MLE turns out to be**
a spread of measurements	normal	the sample mean
yes/no outcomes	Bernoulli	the fraction of yeses
counts per interval	Poisson	the average count
waiting times	exponential	1/the average wait

Example Two: Steps 1 To 3, For A Bernoulli

A concrete cube either fails its strength test or passes. In n tests, k failed.

Step 1. One observation scores θ if it failed, $1 - \theta$ if it passed.

Step 2. Multiply over the n tests.

Step 3. Take logs.



Insight

Same three moves, and the powers collect themselves because every failure contributes a θ and every pass a $1 - \theta$.

Example Two: Step 4, And The Answer

Step 4. Differentiate, set to zero, solve for θ.

Multiply out and collect the θ terms.

Summary

6 of 20 cubes failed $\Rightarrow \widehat{\theta} = 0.3$. Two examples, one method: write $p(x \mid \theta)$, multiply, take logs, find the peak.

Try it: `ml2026.floodresq.com/explore/mle` — the same 6 of 20, and two more distributions, panel 2

Priors, MAP, And Uncertainty

When k/n Is Not Usable

A small survey

Three cubes tested. None failed. So $n = 3$, $k = 0$, and maximum likelihood says $\widehat{\theta} = 0/3 = 0$.

Warning

That answer says failure never happens. Three tests are not enough to support it.

Insight

Maximum likelihood uses the data and nothing else. With little data the peak can sit on the edge, and the edge is a claim we would not make.

Writing Down What You Knew Before

Put a curve on θ before looking at data. For a proportion, use

$$p(\theta) \;\propto\; \theta^{\,a-1}(1-\theta)^{\,b-1} \qquad \text{(the \textbf{Beta} distribution).}$$

Multiply the prior by the likelihood $\theta^{k}(1-\theta)^{\,n-k}$.

Add the exponents.

Insight

The result is a Beta again, with $a \to a+k$ and $b \to b+n-k$. A prior that comes back in the same form is called **conjugate**. And look at *where* the prior's numbers landed — in the exponents, alongside the data's counts:

	power of θ	**power of** $1-\theta$
from the data	k	$n-k$
from the prior	$a-1$	$b-1$

MLE, MAP, And How Sure We Are

The prior sat in the exponents next to k and $n-k$, so it acts as $a-1$ extra failures and $b-1$ extra passes. With $a=b=2$ that is **one extra of each**, and the peak becomes

$$\widehat{\theta}_{\text{MAP}} = \frac{k+a-1}{n+a+b-2} = \frac{k+1}{n+2}.$$

Data	MLE k/n	MAP $(k{+}1)/(n{+}2)$	spread of θ
0 of 3	0.00	0.20	± 0.16
30 of 100	0.30	0.30	± 0.05

Where the last column comes from. The posterior is $\mathrm{Beta}(\alpha, \beta)$ with $\alpha = a + k$ and $\beta = b + n - k$; the spread is its standard deviation, which is the familiar $\sqrt{\bar{\theta}(1-\bar{\theta})/n}$ with the prior's counts included:

$$\text{spread} = \sqrt{\frac{\bar{\theta}\,(1-\bar{\theta})}{\alpha+\beta+1}}, \qquad \bar{\theta} = \frac{\alpha}{\alpha+\beta} \ \text{(the posterior } \textit{mean}\text{)}.$$

Insight

With little data the prior dominates, and MAP does not return zero. With plenty of data the two agree. The last column is what MLE does not report: **how uncertain θ is**, which is the width of the posterior rather than its peak.

Try it: `ml2026.floodresq.com/explore/mle` — watch the posterior move as tests arrive, panel 3

What We Just Assumed

Assumption	Where it entered	When it breaks
independence	the multiplication in Step 2	sensors in a row, days in a storm
the shape is right	writing $p(x \mid \theta)$ at all	a bell fitted to lopsided data
one best answer	setting the derivative to zero	several peaks, or a flat ridge

Warning

Maximum likelihood answers "which θ makes my data least surprising". It does *not* say the model is right, and it hands back **one number with no uncertainty attached**.

What Today Gives You

$$\underbrace{p(x_i \mid \theta)}_{\text{one reading}} \;\to\; \underbrace{\prod_i}_{\text{independence}} \;\to\; \underbrace{\sum_i \log}_{\text{logs}} \;\to\; \underbrace{\frac{d}{d\theta} = 0}_{\text{maximise}} \;\to\; \widehat{\theta}$$

Summary

These four steps do not depend on which distribution you chose. Next lecture applies them to a straight line's prediction, and the result is $(y - \widehat{y})^2$.

Question

Exit check: a road survey finds potholes on 7 of 28 sampled segments. Write the likelihood, and give $\widehat{\theta}$.